

From Ground to Air, From Point to Area: Integrating AI Drones into Advanced Surveying Technologies



STRONG
Engineering Consulting Co.,Ltd. / Aviation Co.,Ltd.



Presenter : Mr. Tsai Sin Da, Licensed Surveyor

Date: June 24, 2026

Presenter's profile

Sinda Tsai

- National Chengchi University –Department of Geomatics
- Head of oversea department, Strong Engineering Co.
- Licensed Surveyor
- Registered APEC Engineer of surveying
- Specialized in mobile mapping system



- Founded in 1984
- 120+ employees

About Strong Co.

- One of the top surveying company in Taiwan.
- Providing high-quality and professional surveying service to our clients.
- All properties including the sensors and platforms are company-owned.
- Expanded our business to East South Asia countries such as Malaysia, East Timor and Indonesia.
- Looking forward to be the pioneer of surveying.



Our Service



Land Surveying



Aerial Surveying



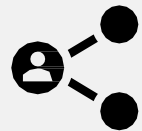
Hydrographic
Surveying



Drone Surveying
& Development

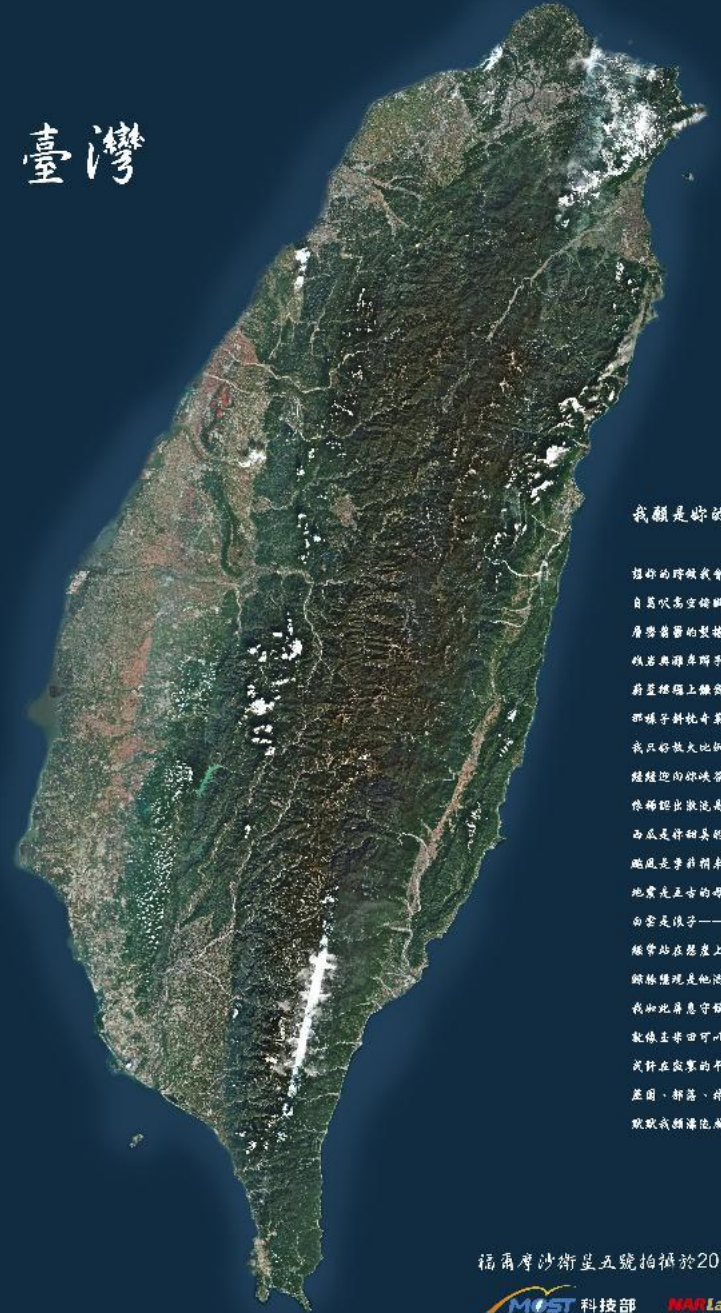
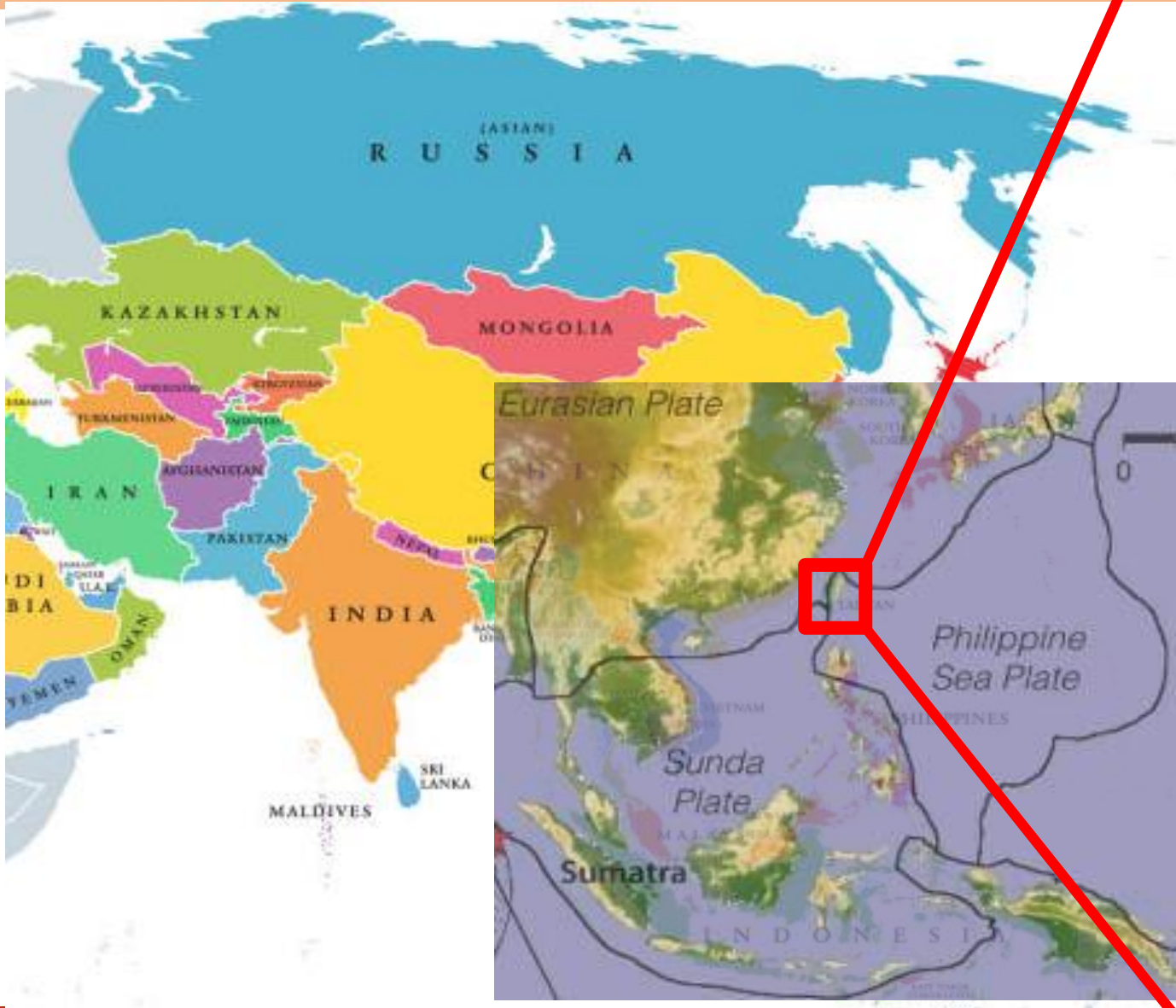


MMS



3D GIS

Where are we from?



我願是你的風景 羅葉

想你的時候我會從衛星地圖
自駕以高空俯瞰你躺臥於島嶼邊緣
磨磨蹭蹭的裝束，橫街斜街的陣眼
礁石與礁石際于旁於交錯網羅絲
蔚藍標圖上離奇穿浪白雷絲達
那樣子斜斜比肩，看不出是礁或礁
我只好放大比例尺覆蓋放下深
絲縫迎向缺缺分明那幽深黑
你稱詔出就是是萬眾，巨石是礁呀
而風是許甜美的鯨白隨礁驚在沙河岸
颶風是幸許朝來的雷信，黑潮是礁絲
地震是古老的母親隨而輕搖你的爭境
白雲是浪子——你像白鷗來的愛情
經常站在懸崖上遙眺海面
靜絲礁現是他沉落異鄉的倒影
我如此屏息守衛而寧靜時未曾覺覺
軟條玉柔回可以不斷斷回四絲絲
莊園在寂寥的平緩，莊園在熱鬧繁榮
莊園、都督、府治、峭壁、神鐘巖
默默我翻滾隨處成你份途的風雪

福爾摩沙衛星五號拍攝於2017年9月至12月

WE SURVEY ANYWHERE





SKY

Optical / Magnetism

- Carrier : Aircraft or drone
- Sensors : LiDAR/Camera/Thermal/Multi spectrum, Magnetic
- Products : Digital terrain model

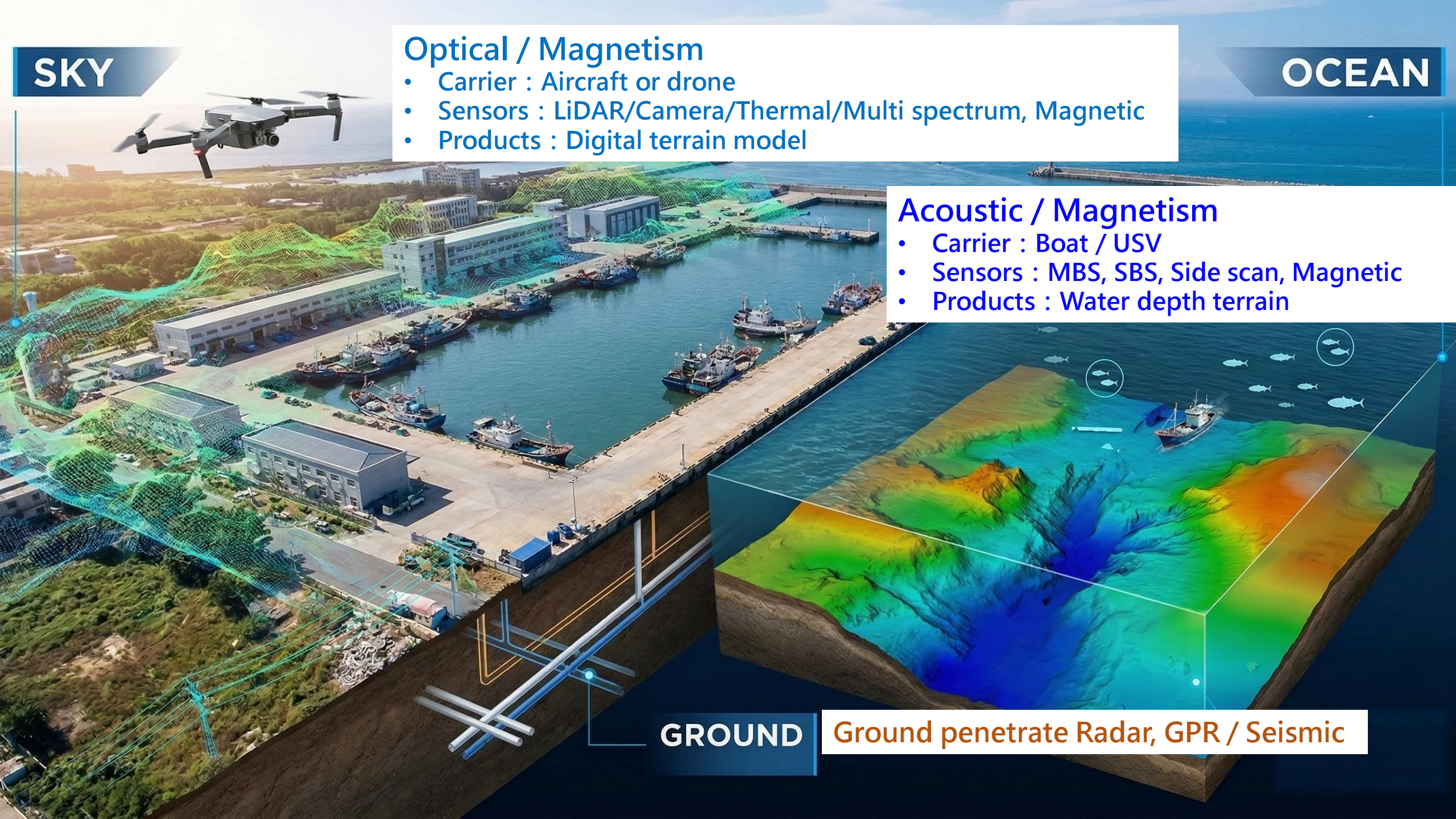
OCEAN

Acoustic / Magnetism

- Carrier : Boat / USV
- Sensors : MBS, SBS, Side scan, Magnetic
- Products : Water depth terrain

GROUND

Ground penetrate Radar, GPR / Seismic



Presentation Outline



Team Capabilities Overview

1



MIT
UAV by
Strong Engineering

2



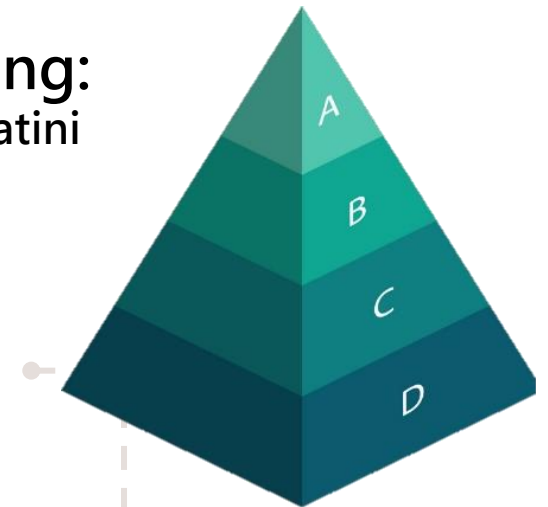
The Brain of the UAV
Flight Control System & Ground
Control Station

3



UAV Surveying:
The Case of Eswatini

Application Focus





Taiwan's UAV Industry Development – Platforms and Applications

Primarily In-House Development
with Supplementary Commercial
Procurement



UAV Platform

- ◆ Propulsion Type (Payload / Endurance)
 - Multirotor / Fixed-Wing / Vertical Take off and Landing (VTOL)
 - All-Electric / Hybrid (Electric-Fuel) / Hydrogen-Powered
- ◆ Safety Mechanism
 - Stall Protection, Collision Avoidance, Geofencing, Redundant Power...

Target
Requirements



Sensor Applications

- ◆ Visible Light Imaging
- ◆ Spectral Imaging
- ◆ Thermal Infrared Imaging
- ◆ LiDAR (Laser Scanning)
- ◆ Sonar, Radar, and Others

Customization



Services and Solutions

- ◆ 3D Geospatial Data Platform
- ◆ Topographic Surveying and Investigation
- ◆ Inspection and Monitoring
- ◆ Cloud-Based UTM & GCS



UAV System Overview

Multimedia Filming & Inspection



2kg, Wheelbase 60cm

ST-Y6-Mini

Aerial Surveying & Inspection



AI Computing Unit **NVIDIA**
12kg, Wheelbase 90cm

ST-Y6-Plus

High Payload Capacity



24kg, Wheelbase 120cm

ST-Y6-Pro

Payload Capacity

0.5

Compact & Portable

1

Multi-Purpose

3

Logistics Operations

5+

kg

Small-Area Surveillance and Defense



ST-X4

4.5kg, Wheelbase 68cm

Aerial Surveying & Inspection



ST-X6120

12kg, Wheelbase 120cm

High Payload & Long Endurance



STH - Military Unmanned Helicopter System

25kg,
Drone Wheelbase
(Motor-to-Motor Distance)
210cm

Advanced
Certification





AI-Powered Autonomous Driving System

■ Core Architecture: Built on Open-Source and Compatible with Proprietary Systems

- Base on Pixhawk Autopilot Bus Specification

pixhawk

ARDUPILOT

PX4
autopilot



■ Avionics Features: Triple-Redundant Communication and Power Systems

- Triple Redundant Digital Power Module Inputs
- Triple Telemetry Ports with Flow Control
- Full GPS Plus Safety Switch Port



- Manufactured and Maintained in Taiwan



■ AI-Powered: NVIDIA Ecosystem

- Compatible with NVIDIA Jetson Orin NX/Nano Series



■ Enhanced Development Flexibility for Integrated Systems

- Best total solution for any type of GCS



■ Remote Piloting and Fleet Management



■ Ensured non-Red supply chain

- NDAA / Blue UAS Ready



■ Supports Tactical Situational Awareness Platform Protocols

- Support Anduril Lattice & TAK (MAVLink Bridge)



■ Basic Specifications

Specifications:

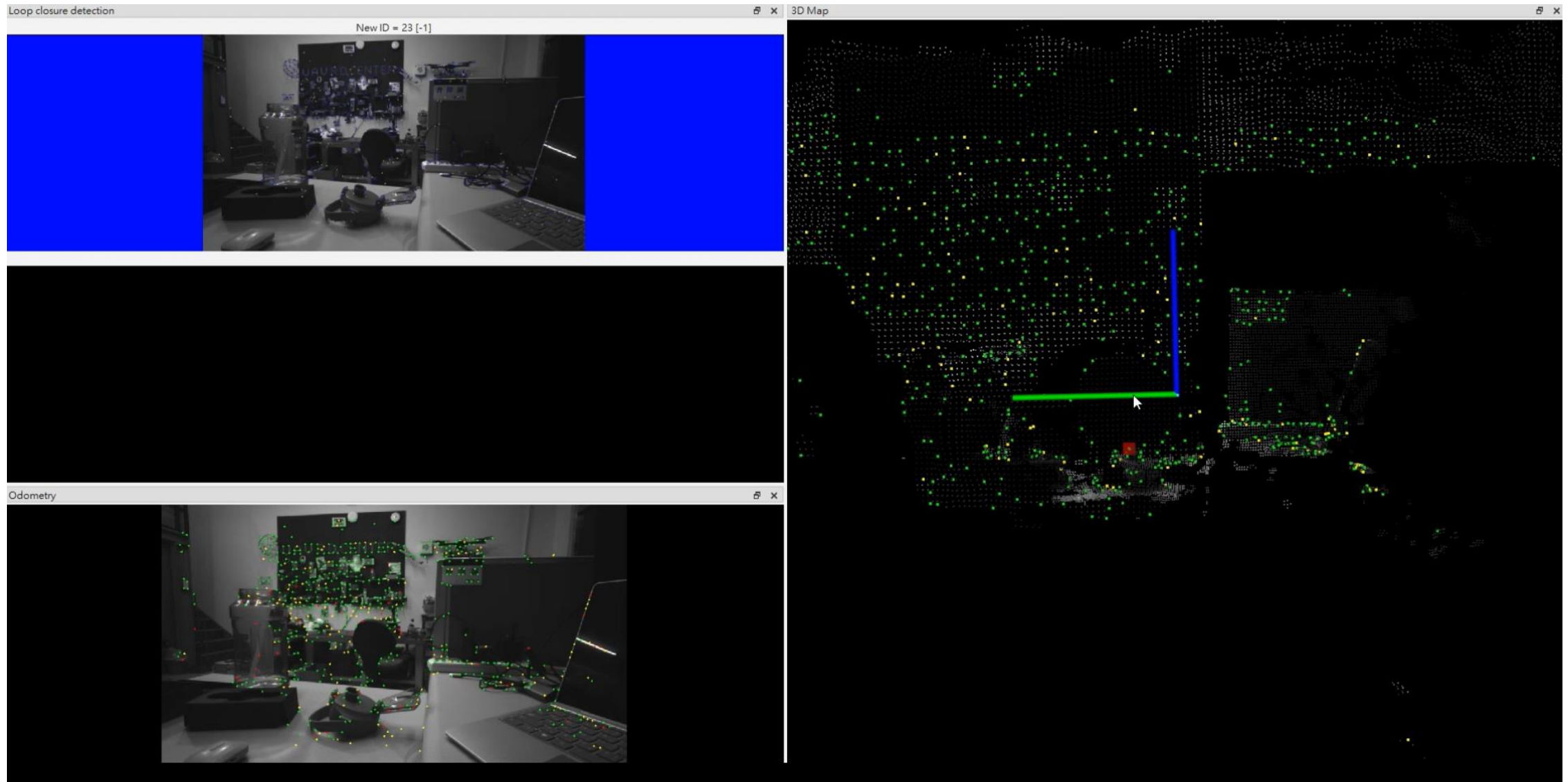
- Size: 133mm x 64mm x 20mm
- Weight: 80g (without Jetson or other modules)

Connections:

- Gigabit Ethernet / MIPI CSI / USB 3
- M.2 Key M 2242 / M.2 Key E 2230
- UART / Mini DisplayPort / CAN Port
- SPI / I2C / I2S Port
- 8 PWM Outputs / RC Port



Indoor SLAM Algorithm Testing (Based on Stereo Vision Sensing)



AI Object Tracking (General Model) – Integrated Gimbal Control





Ground Control System (GCS) Technology and System Development

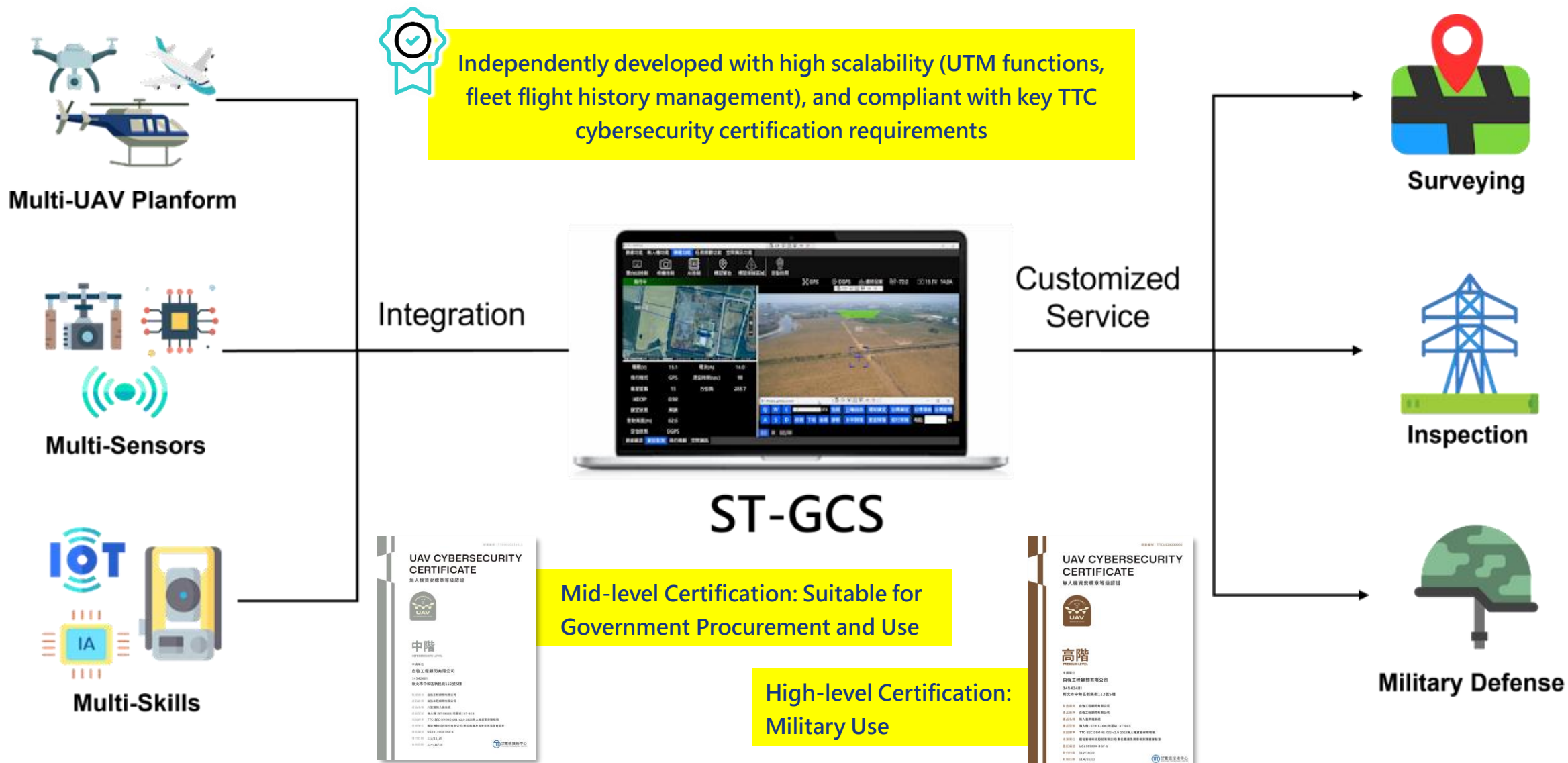




Image Fusion

ST-GCS Function Demonstration

無人機導控站

連線功能 無人機功能 酬載功能 任務規劃功能 除錯工具 **測繪功能**

手繪區域 匯入區域 新增區域點 編輯區域點 建立緩衝區 一般飛行 仿地飛行 任務報表 關閉

可起飛

姿態 10 1.2 RTK(固定) 0.0V 12.6V 0.0A 99%

航拍名稱: 航線規劃21

< 區域點 01 > (請逆或順時針新增)

選擇相機: ILX-LR1(35mm)

GSD : 1.3 cm/pix

航線高度: 120.0 m

-10 -5 -1 +1 +5 +10

航線速度: 5.0 m/s

-5 -1 +1 +5

航線角度: 0.0 度

-50 -5 -1 +1 +5 +50

2024/4/3 下午 06:03:35

航向重疊率: 60.0 %

2024/4/3 下午 06:03:37

2024/4/3 下午 06:03:40

2024/4/3 下午 06:03:40

52.09 Meters

自強工程顧問有限公司@研發部門

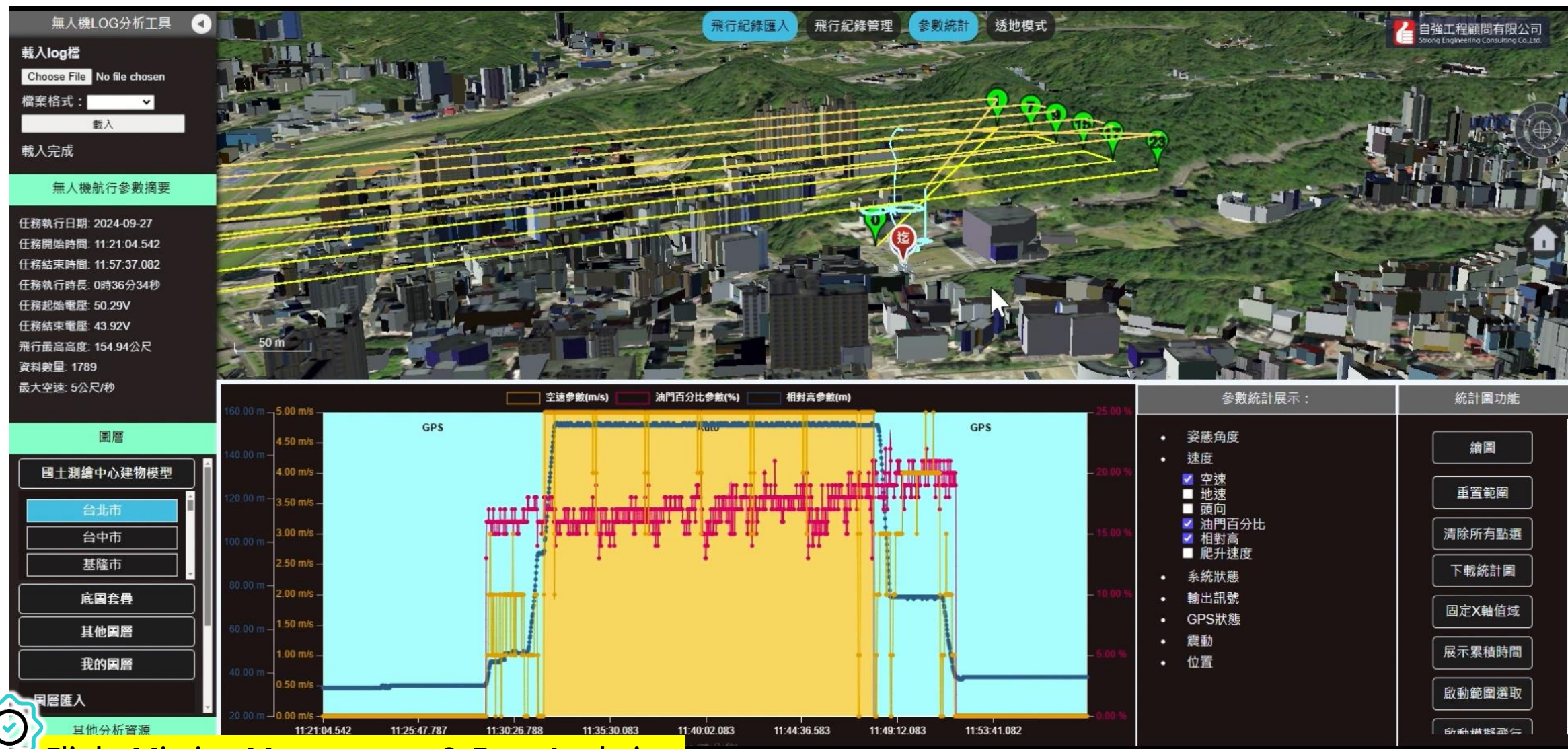
開機自檢 資訊監測 飛行規劃

Buffering 25°07'28.95" N 121°26'40.46" E Alt: 760.64 Meter AGL Dir: 1.50

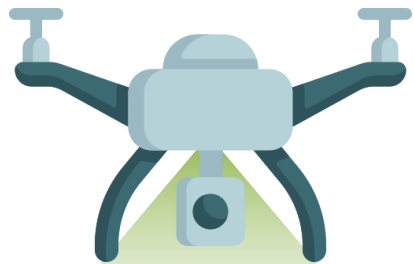
仿地飛行

Windows 11 taskbar with search bar and various application icons.

Communication, Control Architecture, and Cloud Systems



Flight Mission Management & Data Analytics



End-to-End Airborne LiDAR System

- In-house Developed UAV Systems
- Comprehensive Airborne LiDAR Technology



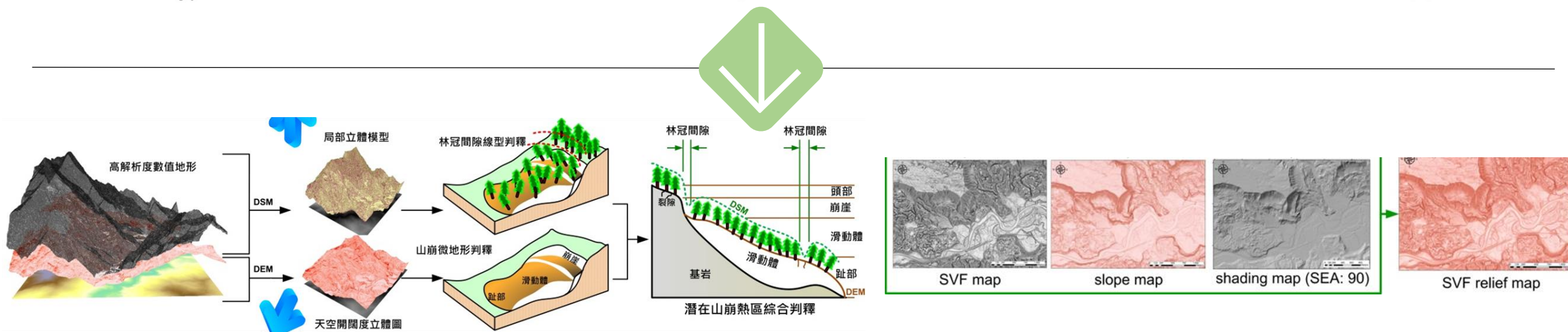
Digital Terrain Data Generation and Quality Control

- Over 10 Years of Experience in National-Scale Data Processing and Quality Control



Multi-Hazard Risk Analysis

- Integrated Team with Multi-Sensor Monitoring Technologies
- Value-Added Data Applications



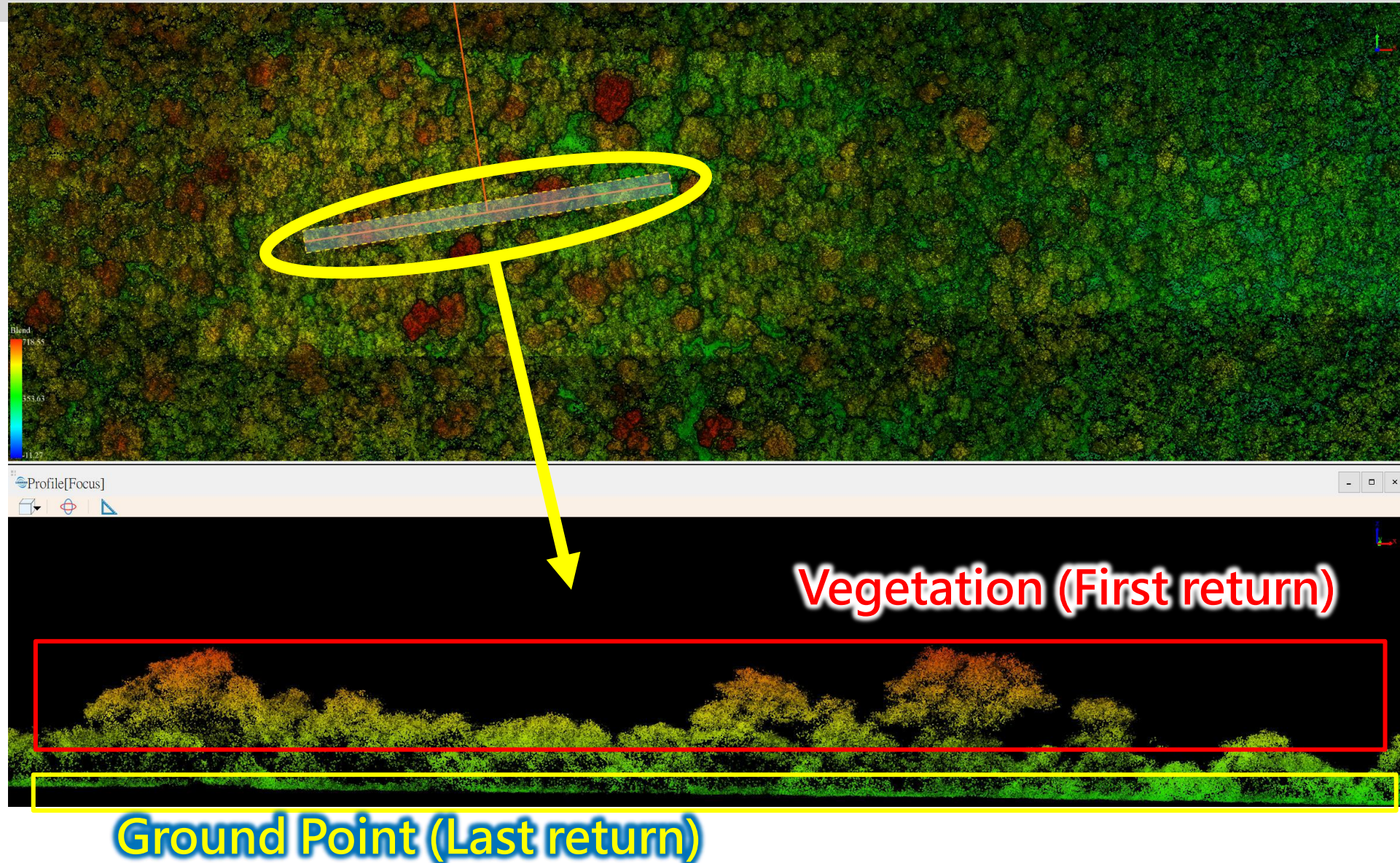


Landscape of the site from drone's view



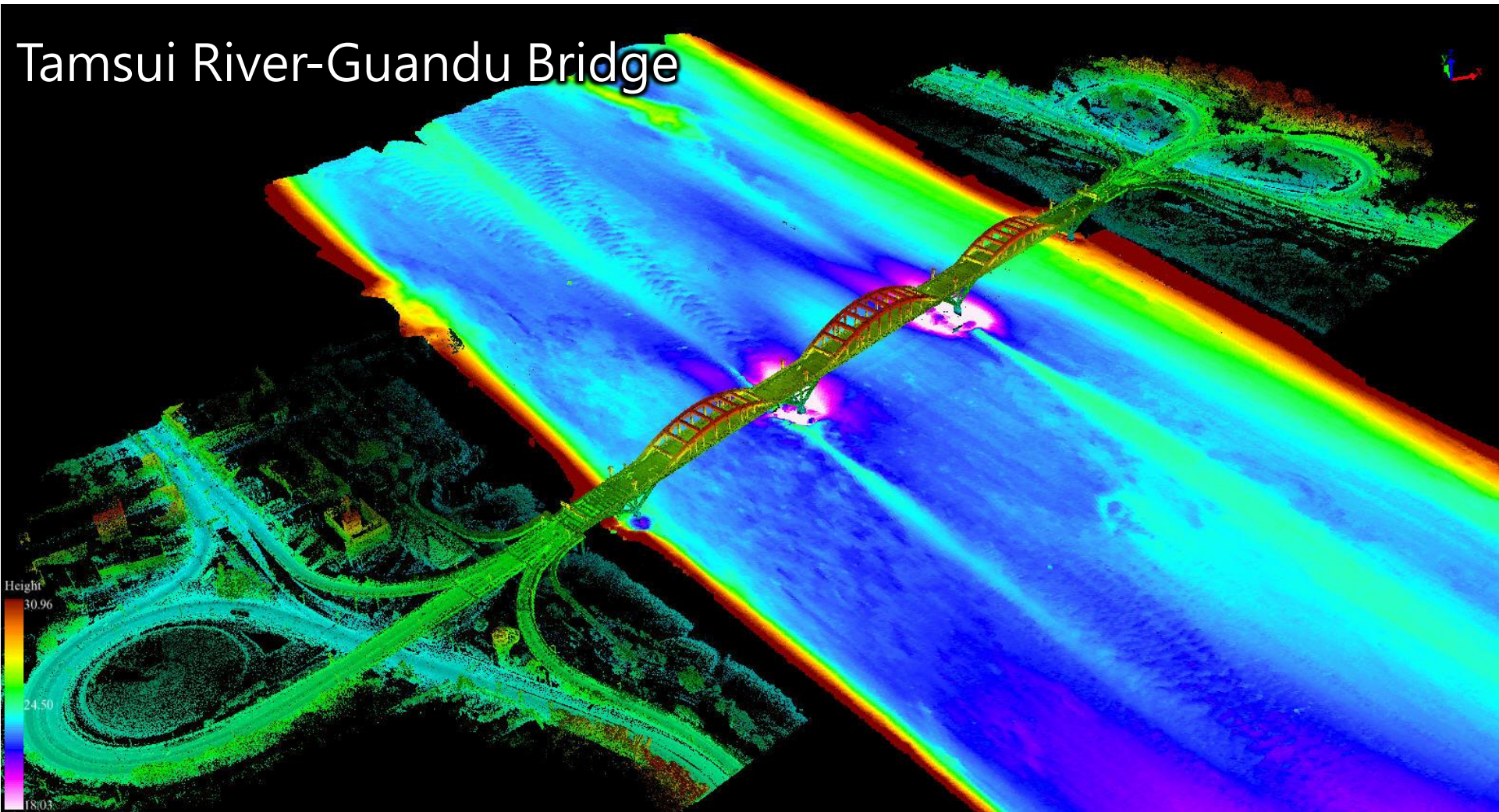
■ ULS surveying in ZA(Eswatini)

Point Cloud Profile Analysis



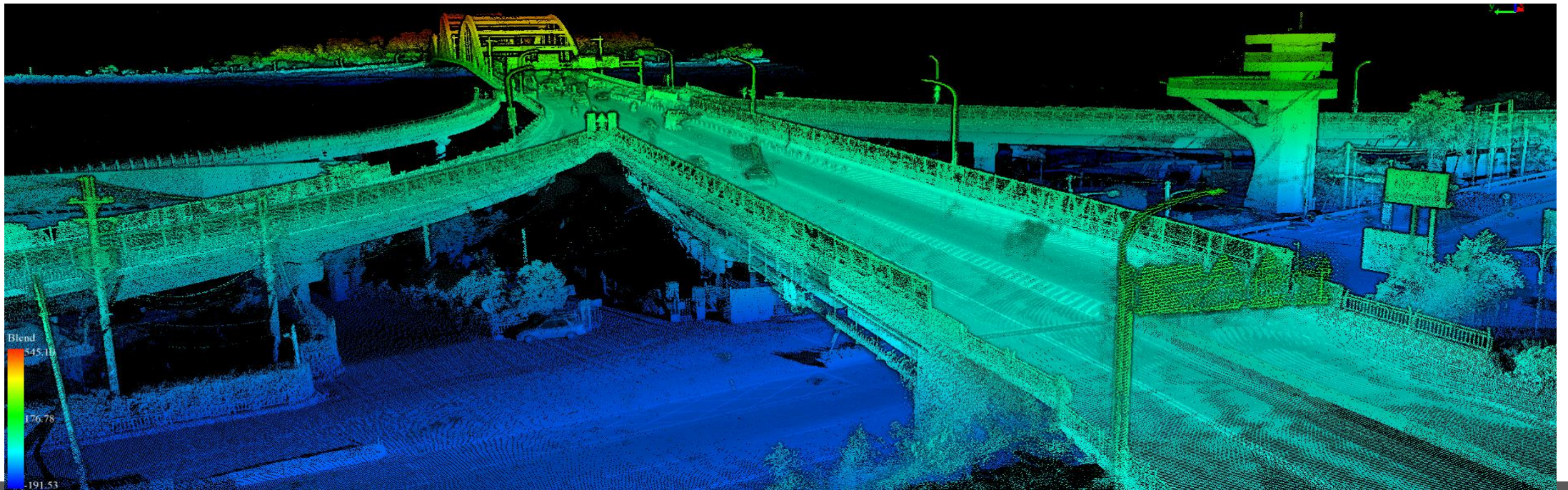
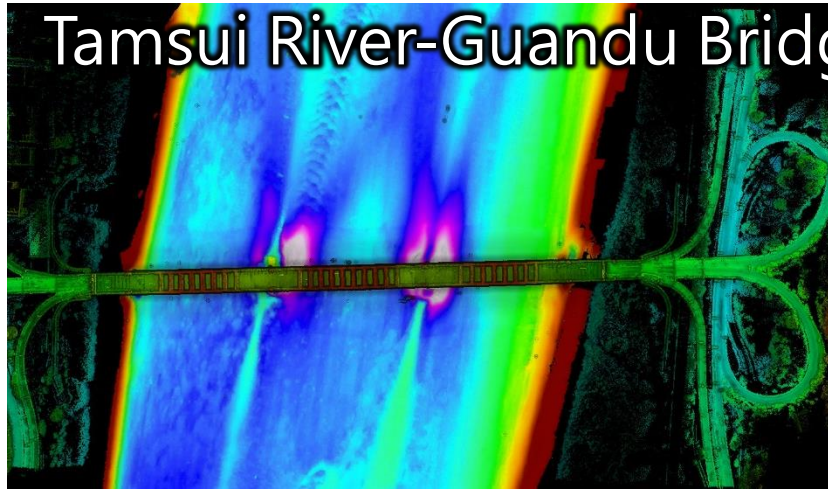
Mobile mapping data integration

■ Vessel LS + ■ Mobile LS + ■ Multibeam Echo Sounder



Scanner on mobile

Tamsui River-Guandu Bridge





Core Technology Development Strategy for Unmanned Systems



Key Subsystems

- ◆ Inertial Navigation / Wireless Communication Systems
- ◆ Controllers / Control Computers
- ◆ Edge AI Computers / Onboard Computers
- ◆ Payloads / Sensor Systems



Taiwan Supply Chain

- ◆ Military / Commercial → Dual-Use Applications
- ◆ Exclusion of Red Supply Chain
- ◆ High Product Development Costs
- ◆ Low-Volume Customization with High Reliability Requirements



Solutions

- ◆ 3D Spatial Data Platform
- ◆ Cloud-based UTM and Fleet Management
- ◆ Mission-Oriented GCS (Ground Control System)
- ◆ UAV System Cybersecurity



自強工程顧問有限公司
Strong Engineering Consulting Co.,Ltd.

Thank You For Your Attention

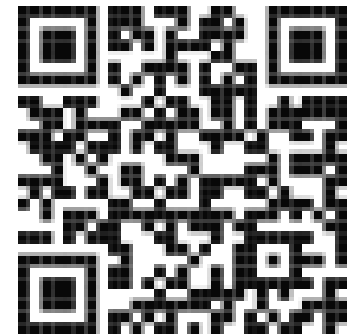
Scan the QR code for more information



Official Website



Youtube



Facebook Page

Multi-Dimensional Spatial Information Base Map Surveying and Updating Plan